

Faisal Farhan

Email: faisalfarhan.eee@aust.edu | LinkedIn: [faisal-farhan](#) | Website: f-farhan.github.io
University of Girona, Spain.

EDUCATION

M.Sc in Medical Imaging and Applications

Europe

Erasmus Mundus Joint Master's Degree

September 2023 - June 2025

Université de Bourgogne, France; University of Cassino, Italy; University of Girona, Spain;

Thesis: Design and Validation of Multimodal and Lightweight Handheld ST-rPPG Acquisition Systems for Physiological measurements.

B.Sc. in Electrical and Electronic Engineering

Dhaka, Bangladesh

Ahsanullah University of Science & Technology

April 2016 - January 2021

GPA: 3.936/4.0; Rank: 1/196

Undergraduate Thesis: Study of Pattern Recognition System through the development of a Raspberry Pi based speaker recognizer.

RESEARCH INTERESTS

Signal Processing, Medical Imaging, Image Processing, Machine Learning & Applied Artificial Intelligence

WORK EXPERIENCES

Graduate Research Intern

Laboratoire en Imagerie et Vision Artificielle (ImVIA), Dijon, France

February 2025-June 2025

Lecturer

Department of Electrical and Electronic Engineering, Ahsanullah University of Science & Technology, Bangladesh.

June 2021 - August 2023

- Prepared and delivered lectures, tutorials, workshops and seminars.
- Developed different curriculum and course materials.
- Conducted research and writing papers, proposals, journal articles.
- Attended and participated in meetings, conferences, and other events in and outside institution.
- Supervised undergraduate students.

Faculty Member

May 2021 - July 2021

School of Science, Engineering & Technology, East Delta University, Bangladesh.

- Prepared and delivered lectures, tutorials, workshops and seminars.
- Attended and participated in meetings, conferences, and other events in and outside institution.
- Supervised undergraduate students.

PUBLICATIONS

1. Kabir, M. S. A., **Farhan, F.**, Siddiquee, A. A., Baroi, O. L., & Rahimi, J. (2023). "Effect of Input Channel Reduction on EEG Seizure Detection. *Przegląd Elektrotechniczny*", 2023(12). [\[Paper Link\]](#)
2. Thahmidul I. Nafi, Erfanul Haque, **Faisal Farhan**, Asif Rahman "High Accuracy Swin Transformers for Image-based Wafer Map Defect Detection", *International Journal of Engineering and Manufacturing*. [\[Paper Link\]](#)
3. **F. Farhan** and T. I. Nafi, "ANN based approach to predict criminal trends in Bangladesh", 2nd International Conference on Artificial Intelligence and Signal Processing (AISP), 2022. [\[Paper Link\]](#)
4. T. H. Rafi, R. M. Shubair, **F. Farhan**, M. Z. Hoque and F. M. Quayyum, "Recent Advances in Computer-Aided Medical Diagnosis Using Machine Learning Algorithms with Optimization Techniques", in *IEEE Access*. [\[Paper Link\]](#)
5. T. H. Rafi, **F. Farhan** (2021), "A Performance Evaluation of Machine Learning Models on Human Activity Identification (HAI)", 12th International Conference on Soft Computing and Pattern Recognition, Springer. [\[Paper Link\]](#)
6. T. H. Rafi, **F. Farhan**, Z. Hoque, F. Quayyum. (2020). "Electroencephalogram (EEG) brainwave signal-based emotion recognition using extreme gradient boosting algorithm", *Annals of Engineering*, 1(1), 0005. [\[Paper Link\]](#)
7. "Revival of the Lost Glory of Muslin through Identification of Phuti Karpas Plant with an Effective Approach by Using Convolution Neural Network", *Smart Agricultural Technology*. [Under Review]
8. "A Dual-Pipeline Approach for Colorectal Polyp Detection and Segmentation with Machine and Deep Learning Models", *Engineering Research Express*. [Under Review]

SELECTED ACADEMIC PROJECTS

- Gastrointestinal Polyp Segmentation and Classification (Advanced Image Analysis, Machine Learning and Deep Learning academic project) [\[pdf\]](#)
- Financial Data Analysis: Predictive Analytics for Trading Actions (Statistical Learning and Data Mining academic project)
- Lung CT Volume Registration (Medical Image Registration academic project) [\[pdf\]](#)
- Brain MRI tissue segmentation (Medical Image Segmentation academic project) [\[pdf\]](#)
- Smart Chest Xray Analysis and Report Generation system (Software Engineering academic project) [\[github\]](#)
- Raspberry Pi based Speaker recognition system (Undergraduate thesis project)

TECHNICAL SKILLS

- Programming Languages: C++, Python, Java, Matlab, Verilog, HTML.
- Libraries: Pandas, NumPy, Matplotlib, Tkinter.
- Data analysis and visualization: Scikit learn, tensorflow, keras, pytorch, matplotlib, excel.
- Tools: Google Colab, Git, Visual Studio, Codeblocks, MATLAB, Arduino.
- Simulators: Orcad Pspice, Quartus, Proteus, Cadence.
- Design Software: Adobe Photoshop, Adobe Illustrator, Figma
- OS: Windows, Linux, Raspbian.

ACADEMIC HONORS

- Erasmus Mundus Scholarship, European Union.
- Aquisis International Mobility Grant, Region Bourgogne Franche Comte (France).
- Dean's List of Honor in all four academic years.
- Khan Bahadur Ahsanullah award (highest CGPA among all department in the graduating class, to be conferred in Convocation'2025)
- Nominee for Prime minister gold medal award (Academic year 2021)

PRESENTATIONS

- ◆ Detection of diabetic retinopathy using a fusion of textural and ridgelet features of retinal images and sequential minimal optimization classifier, *University of Burgundy (France)*.
- ◆ Review on Traditional and Deep Learning Methods for Diabetic Retinopathy detection and classification, *University of Burgundy (France)*. 2022
- ◆ 2nd IEEE International Conference on Artificial Intelligence and Signal Processing ([AISP](#)).
- ◆ 12th International Conference on Soft Computing and Pattern Recognition ([SOCPAR'21](#)). 2020

LEADERSHIP EXPERIENCES

- **Member**, Erasmus Mundus Association 2023-2025
- **Treasurer**, AUST Career Development Club 2022-2023
- **Chittagong Division Representative**, Bangladesh Development Project 2021
- **Vice President**, Ahsanullah University of Science & Technology EEE Society 2020
- **Lead Volunteer**, Youth's Voice Bangladesh 2013-2015